1. MOTIVATION. GROUPS AS METRIC SPACES

I. Introduction: Path-metrics and Cayley Graphs

$(G,\cdot)$ $\longleftrightarrow$ (G, d_S)

Algebraic object — Metric space

Algebraic features $\longleftrightarrow$ (Coarse) geometric features

* G f.g. group. Let S be a finite generating set.
  Assume S is symmetric ($s \in S \Rightarrow s^{-1} \in S$).
  $\Gamma = Cay(G,S)$;
  - Vertex set of Γ is the underlying set of G
  - If $g \neq h \in G$, $\{g, h\}$ is an edge if $\exists s \in S$ such that $g = hs$.

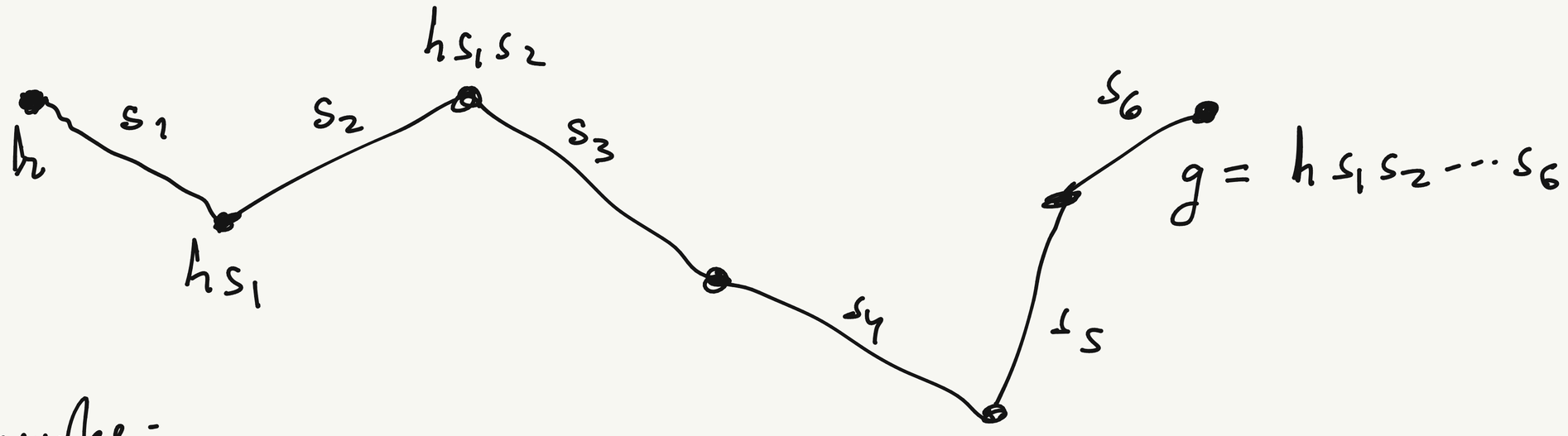
* Given a simple (simplicial) graph $\Gamma = (V, E)$ connected,
  V: vertex set; E: unordered pairs of vertices

  $d_\Gamma : V \times V \to \mathbb{R}$;
  $(x,y) \mapsto d_\Gamma(x,y) = \min \{ \text{len}\, \gamma \mid \gamma \text{ is a path in } \Gamma \text{ from } x \text{ to } y \}$

* Given a f.g. group G and a sym. fin. generating set S,
  $(G,S) \rightsquigarrow Cay(G,S) \rightsquigarrow (G, d_S)$
  conn. graph

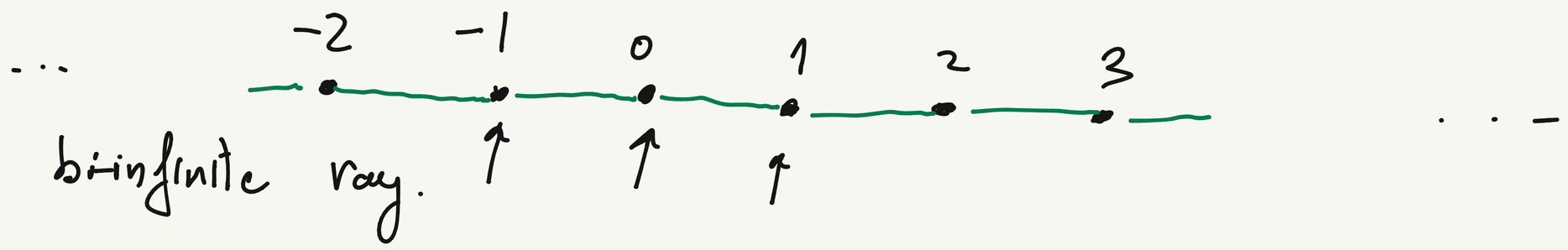
  $d_S : G \times G \to \mathbb{R}$;
  $d(g,h) = \underbrace{\| g^{-1} h \|}_{\text{word length}} =$

  $= \min \{ n \in \mathbb{R}_{\geq 0} \mid \exists s_1, \dots, s_n \in S \text{ such that } g = h s_1 s_2 \cdots s_n \}$



- Examples:

i) $\mathbb{Z} =: G$, $S = \{\pm 1\}$

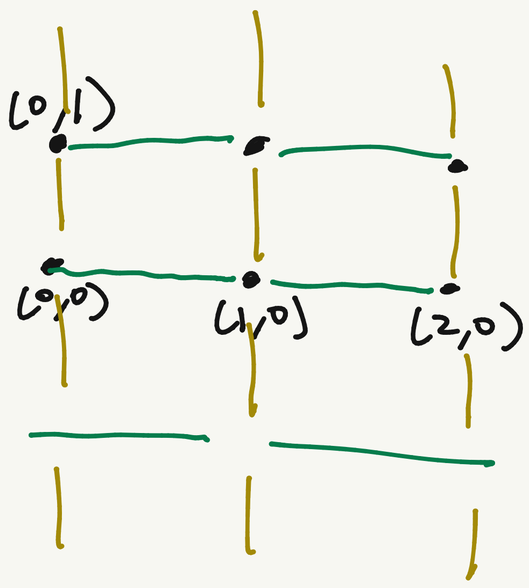


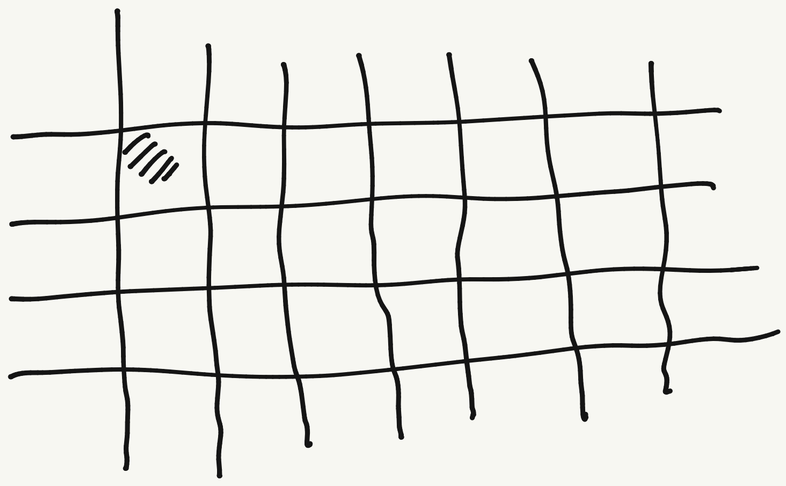
$\mathbb{Z} \hookrightarrow \mathbb{R}$;

$\mathbb{Z} \sim \mathbb{R}$

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ii) $\mathbb{Z}^2 = \mathbb{Z} \times \mathbb{Z}$; $S = \{\pm e_1, \pm e_2\}$ $[e_1 = (1,0)]$ $[e_2 = (0,1)]$

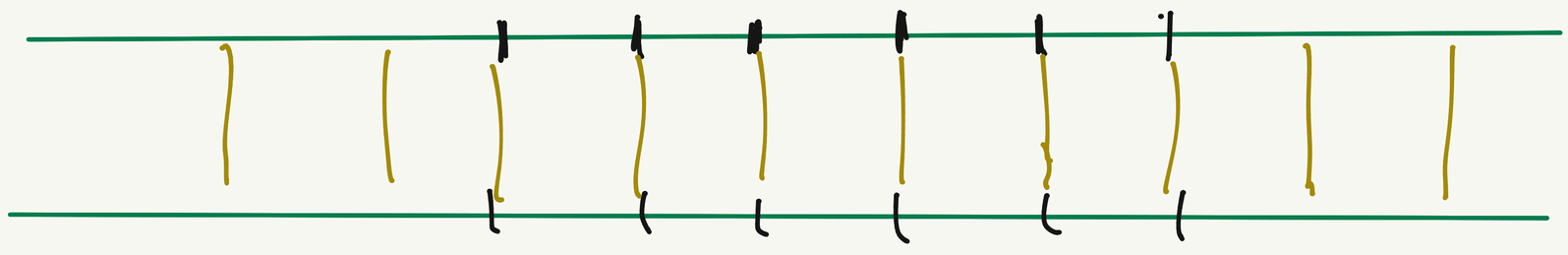


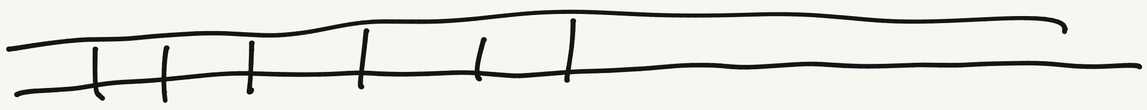




$\mathbb{Z}^2 \sim \mathbb{R}^2$

iii) $\mathbb{Z} \times \mathbb{Z}/2$; $S = \{\pm e_1, e_2\}$; $e_1 = (1,0)$, $e_2 = (0,1)$, $1+1 = 0$



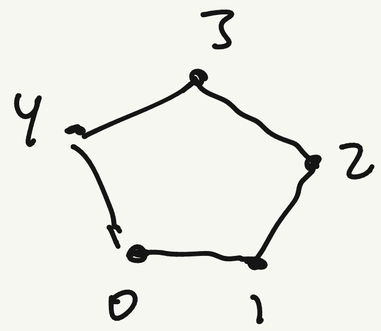


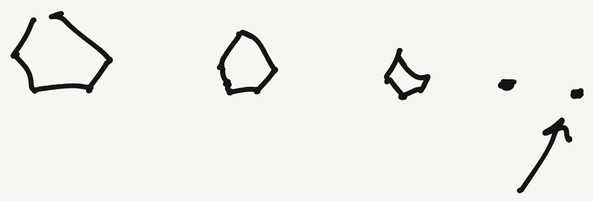
$\mathbb{Z} \times \mathbb{Z}/2 \sim \mathbb{R}$



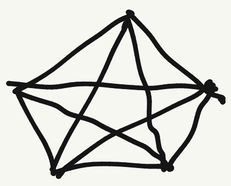
$\mathbb{R}$

iv) $\mathbb{Z}/5$, $S = \{\pm 1\}$;





$S = \{\pm 1, \pm 2\}$;

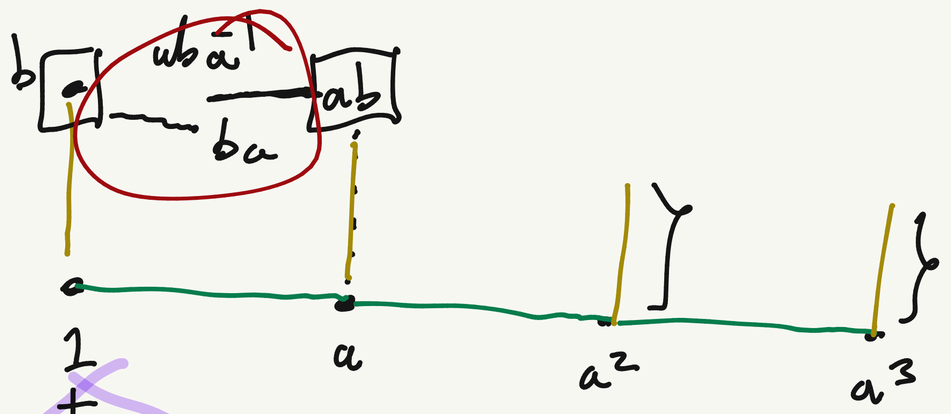


$\sim \bullet$

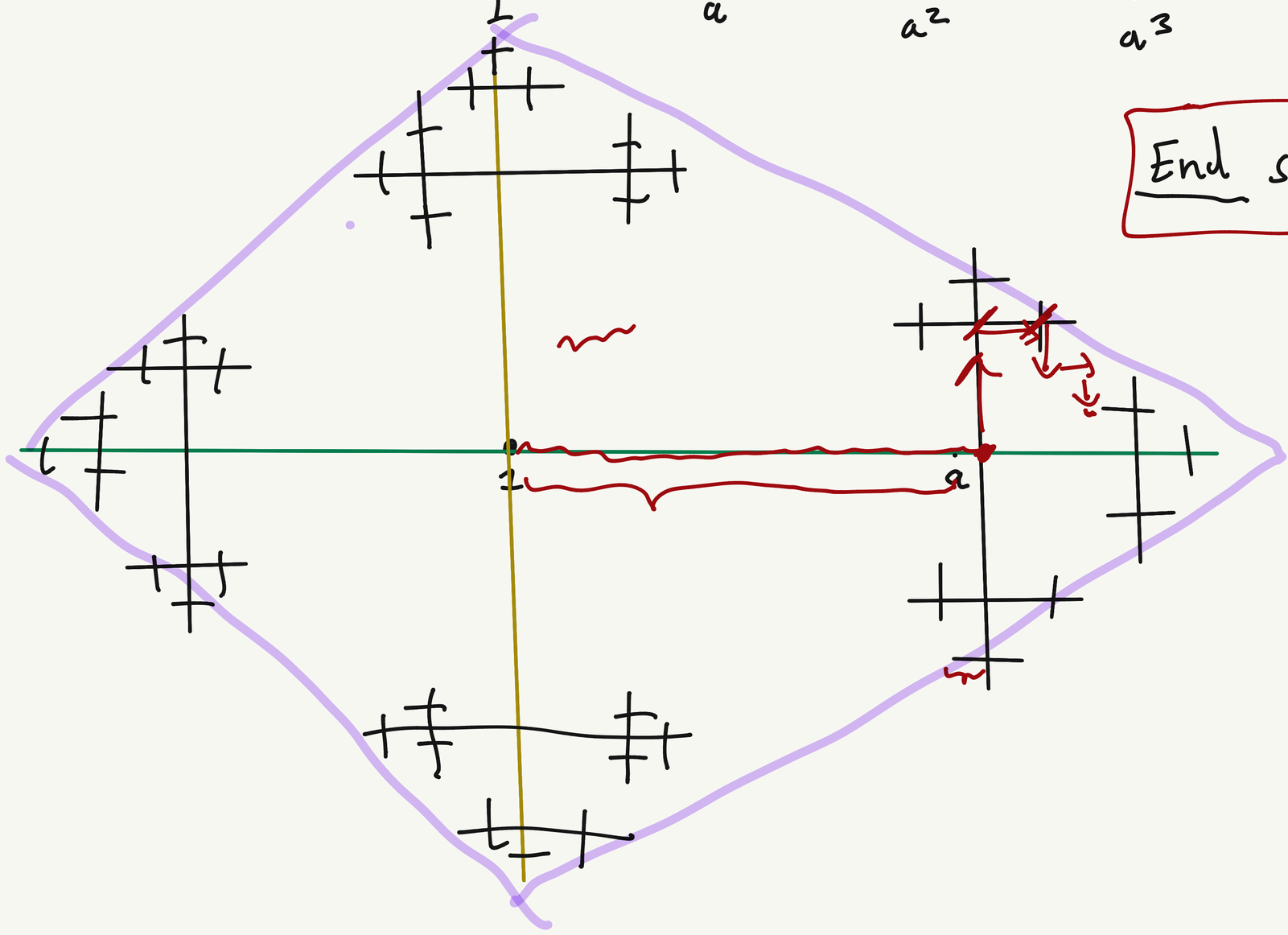
If G is finite, $G \sim *$.

v) $F_2 = \langle a, b \rangle = \mathbb{Z} * \mathbb{Z}$ $\quad S = \{a^{\pm 1}, b^{\pm 1}\}$

$Cay(F_2, S)$



End space



II. Quasi-isometries

- Let $f:(X,d_X) \to (Y,d_Y)$ a map between metric spaces.

  i) Let $c>0$, $b \geq 0$. f is said to be a (c/b)-quasi-isometric embedding if

$$\frac{1}{c} d_X(x,x') - b \leq d_Y(f(x), f(x')) \leq c \cdot d_X(x,x') + b$$

  ii) f is a quasi-isometric embedding (q-i. emb.) if such c/b exist.

Let $f_1, f_2 : X \to Y$. f_1, f_2 are said to be "close" if

$\exists r \geq 0$ s.t. $d(f_1(x), f_2(x)) \leq r$

  $\therefore$ $f:X \to Y$ is a quasi-isometry (QI) if it is a q-i. emb and admits a q-i. emb. $g:Y \to X$ such that

$$gf \approx id_X \qquad fg \approx id_Y$$

$$\left(\begin{array}{l} f:X \to Y \text{ in } Top \\ g:Y \to X \end{array} \right. , \quad \left. \begin{array}{l} gf \simeq id_X \\ fg \simeq id_Y \end{array} \text{ analogy} \right)$$

- X and Y are quasi-isometric ($X \underset{QI}{\sim} Y$) if $\exists$ a QI between them.

Examples:

i) $\mathbb{Z} =: G, \quad S = \{\pm 1\}$;

$f: \mathbb{Z} \hookrightarrow \mathbb{R}$ is a QI. f is a $(1,0)$-q-i. emb.

$g: \mathbb{R} \longrightarrow \mathbb{Z}$; $y \mapsto \lfloor y \rfloor$ g is a $(1,2)$-q.-i. emb.

$gf = id_{\mathbb{Z}} \simeq id_{\mathbb{Z}}$, $fg \simeq id_{\mathbb{R}}$

$$\mathbb{Z} \underset{QI}{\sim} \mathbb{R}$$

ii) $\mathbb{Z}^2 \hookrightarrow \mathbb{R}^2$ is a QI. In general, $\mathbb{Z}^n \underset{QI}{\sim} \mathbb{R}^n$.

iii) $\mathbb{Z}/3 \underset{QI}{\sim} *$. In general, $G \underset{QI}{\sim} *$ iff G is finite.

Finiteness of f.g. is a QI (geometric) invariant.

iv) $G = \mathbb{Z} \times \mathbb{Z}/2$ i

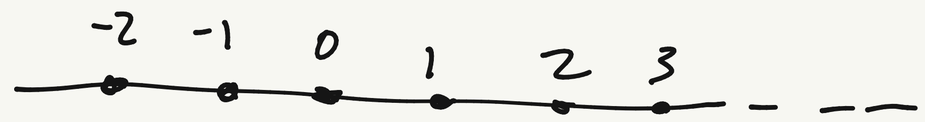
$$\begin{array}{ccc} \mathbb{Z} & \xrightarrow{QI} & \mathbb{Z} \times \mathbb{Z}/2 \\ x & \longmapsto & (x,0) \end{array}$$

v) $F_2 = \langle a, b \rangle$

Dependance of $d_S : G \times G \to \mathbb{R}$ on S.

$$G \text{ f.g.} \dashrightarrow Cay(G,S) \longrightarrow (G, d_S)$$

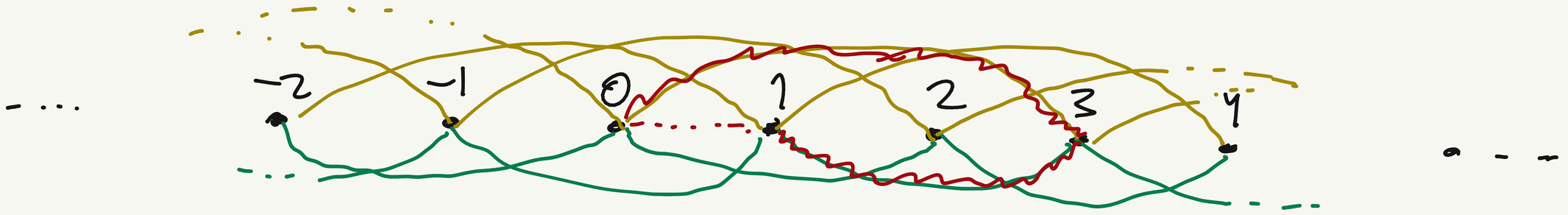
i) $\mathbb{Z} =: G$, $S = \{\pm 1\}$



$d_S(0,1) = 1$

ii) $\mathbb{Z} =: G$, $S' = \{\pm 2, \pm 3\}$

⋄ $n = 6$, $S' = \{\pm 2, \pm 3\}$.



$d_{S'}(0,3) = 1$; $d_{S'}(0,1) \underset{>1}{\overset{\leq 2}{=}} 2$

* **Proposition**: Let G be a f.g. group and S, S' two finite symmetric generating sets. Then

$$f = \text{“}id_G\text{”} : (G, d_{S'}) \longrightarrow (G, d_S)$$

is a QI.

Proof: Given $s' \in S'$, we can write $s' = s_1 \cdots s_{K(s')}$ for $s_1, s_2, \dots, s_{K(s')} \in S$. Let $m_1 \in \mathbb{N}$ such that $K(s') \leq m_1$ $\forall s' \in S'$.

Let $g \in G$. Consider $K = d_{S'}(1, g)$. $g = s_1' \cdots s_K'$ w/ $s_i' \in S'$.
$\longrightarrow$ g is written as a product of at most $K \cdot m_1$ elements of S.

$$d_S(1, g) \leq m_1 \cdot d_{S'}(1, g). \quad (*)$$

For arbitrary $g, h \in G$, $\boxed{d_S(g, h) \leq m_1 \cdot d_{S'}(g, h)}$

$$d_S(g, h) = d_S(1, g^{-1}h) \quad (*)$$

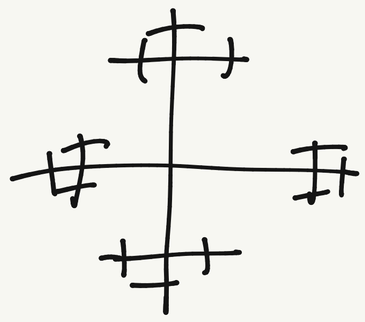
$$\left[\frac{1}{m_2} d_{S'}(g, h) \leq d_S(g, h) \leq m_1 d_{S'}(g, h) \right]$$

If $m = \max\{m_1, m_2\}$, f is a $(m, 0)$-q-i. emb.
f^{-1} is a q-i. emb. $ff^{-1} = id \approx id$, f QI.
$f^{-1}f = id \approx id$

□

- Free groups and their QI-type:

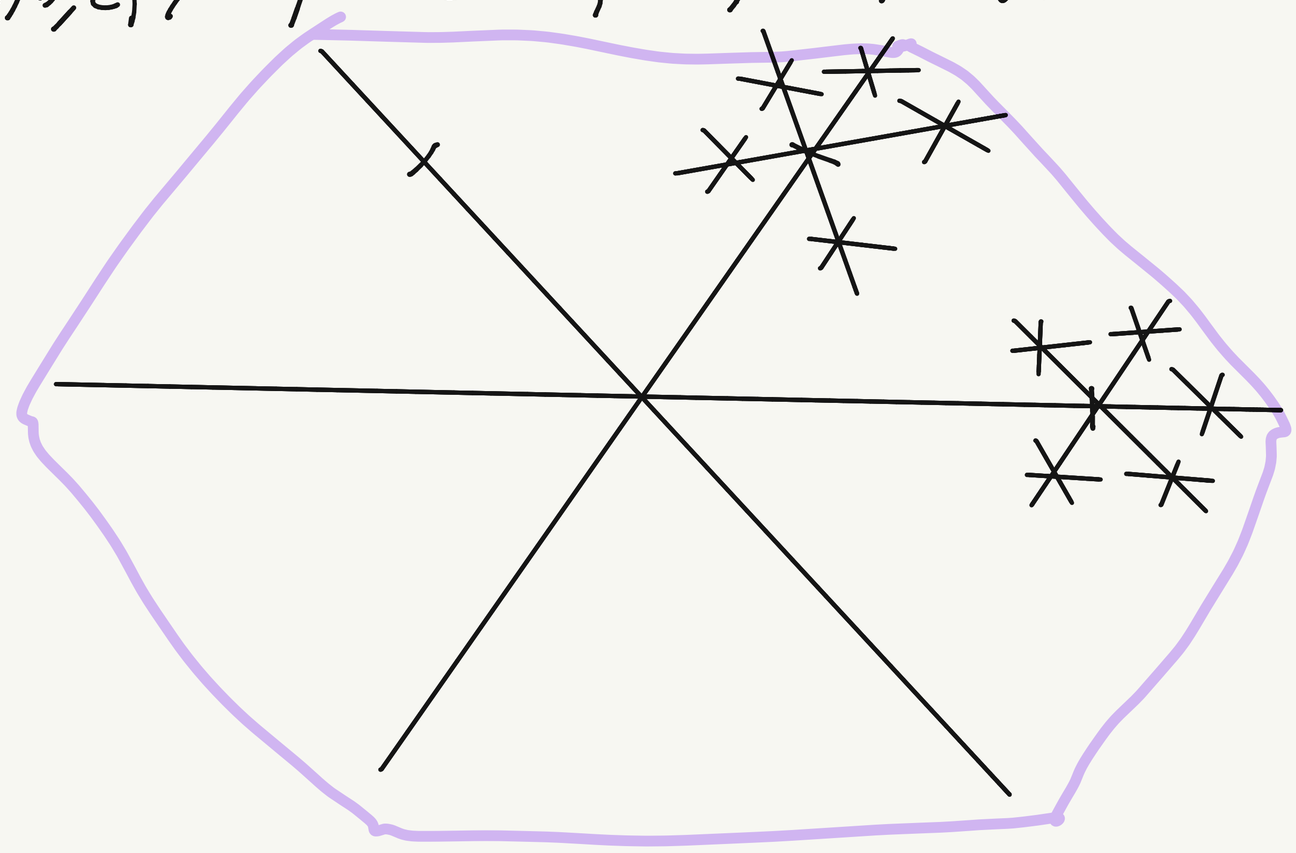
$F_1 = \langle a \mid \rangle = \mathbb{Z}$ $\not\underset{QI}{\sim}$ $F_2 = \langle a, b \mid \rangle$



$F_n = \langle x_1, \dots, x_n \mid \rangle = \mathbb{Z} * \overset{n}{\dots} * \mathbb{Z}$; $n \geq 2$

$F_1 \underset{QI}{\not\sim} \left[F_2 \underset{QI}{\sim} F_n \right] \quad \forall n \geq 2$

$F_3 = \langle a, b, c \mid \rangle$; $S = \{a^{\pm 1}, b^{\pm 1}, c^{\pm 1}\}$



Rank of a group G (min. number of generators a fin. gen. set admits)

Algebraic property

is <u>not</u> a geometric property.

<u>Q</u>

- Consider the group $G = \mathrm{Isom}(\mathbb{Z})$ of isometries of $\mathbb{Z}$

  Find a finite gen. set S of G, draw $\mathrm{Cay}(G,S)$ and recognize its quasi-isometry type.

  $G = D_\infty$ infinite dihedral group.